��������������������

[illegible]

[illegible] `vision_qa_platform` [illegible] Web [illegible]

[illegible]

[illegible] **Flask** [illegible] **+ HTML/JS** [illegible] `webapps_ctl.sh` [illegible]

[illegible]	[illegible]	[illegible]
`trace_webapp`	7860	[illegible] JSON [illegible]
`chat_vqa_webapp`	7861	[illegible] **Demo**[illegible]
`chat_vqa_portal`	7862	[illegible]MongoDB [illegible]

[illegible]

```
flowchart TB
    subgraph Frontend["[illegible]"]
        T7860["7860 [illegible] | D3[illegible] | [illegible]"]
        T7861["7861 [illegible] | D3[illegible]"]
        T7862["7862 [illegible] | [illegible] | D3[illegible]"]
    end

    subgraph Backend["Flask [illegible]"]
        TW["trace_webapp/app.py"]
        CW["chat_vqa_webapp/app.py"]
        CP["chat_vqa_portal/app.py"]
    end

    subgraph Shared["[illegible]"]
        LLaVA["LLaVA-1.5-7B\n[illegible]→[illegible]"]
        Qwen["Qwen2-1.5B\n[illegible]"]
        RM["reasoning_modules.py\n[illegible]/[illegible]/[illegible]/[illegible]"]
    end

    subgraph Data["[illegible]"]
        JSON["modular_results_*.json\n[illegible]"]
        DB["MongoDB / Mongita\n[illegible]+[illegible]+[illegible]"]
    end

    T7860 --> TW
    T7861 --> CW
    T7862 --> CP

    TW --> JSON
    TW --> RM
    CW --> RM
    CP --> RM

    RM --> LLaVA
    RM --> Qwen
    CP --> DB
```

□□□□

```
cd /root/autodl-tmp/vision_qa_platform/agent_workspace
bash webapps_ctl.sh start    # □□□□□□
bash webapps_ctl.sh status   # □□□□
```

□ AutoDL □□□□□□□□□□□□ SSH □□□□□□□□

```
ssh -p 24484 \
  -L 7860:127.0.0.1:7860 \
  -L 7861:127.0.0.1:7861 \
  -L 7862:127.0.0.1:7862 \
  root@connect.bjb1.seetacloud.com
```

□□□□□

- 7860□http://127.0.0.1:7860
- 7861□http://127.0.0.1:7861
- 7862□http://127.0.0.1:7862

□□□□□□□□□□□□□□□□□□

7860 / 7861 / 7862 □□□□□□□□□□□□□□portal □□ `vqa_service.py` □□ webapp □□□□

1. □□□□□LLaVA□

`VisionToTextReasoner`□`tot_got_integration.py`□□ **LLaVA-1.5-7B** □□□□□□□□□□□

- □□□□□`llava-1.5-7b-hf`
- □□□□□`get_visual_description(image_path, question_hint)`
- □□□PIL □□ → CLIP □□□ → LLaVA `generate` → □□ `ASSISTANT:` □□□□□□

2. □□□□□□Qwen2-1.5B□

`SearchController.solve()`□`reasoning_modules.py`□□□□□□□

□□	□	□□
□□□□□	`QuestionDecomposer`	□□□ → 3 □□□□
□□□□□	`ThoughtGenerator`	□□□□□□□ 3 □□□□□□ + □□□
□□□□□	`StateEvaluator`	□□□□□□0.1~1.0□

□□	□	□□
□□□□□	`SearchController`	□□□□□□□□□□□□□□

□□□□□□

```
sub_questions = decomposer.decompose(visual_description,
question)
for sub_q in sub_questions:
    candidates = generator.generate_candidates(...)
    scores = evaluator.score_candidates(...)
    best_idx = argmax(scores)
    trace.append({ sub_question, candidates, scores,
selected_index })
final_answer = aggregate(selected_sub_answers)
```

3. □□□□

`constrained_finalize_answer()`□`vqa_eval_utils.py`□□□□□□□□□□□□□□□□□□□□□□□□□□

4. □□□□ + □□□

□□□□□□ `_PIPELINE` □□□□ + `threading.Lock()`□□□□□□□□□□□ □□□□□□□□□□□ GPU□

□□□□□

- □□□`/root/autodl-tmp/vision_qa_platform/llava-1.5-7b-hf`
- □□□`/root/autodl-tmp/vision_qa_platform/text_model/qwen2_1_5b_instruct`

□□7860□`trace_webapp` — □□□□□□

□□□□□□□□□□□□□□□□ **JSON**□□ `trace` □□□□□□□□□□□□□□□□□□□

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- □□□`agent_workspace/trace_webapp/app.py`
- □□□`agent_workspace/trace_webapp/templates/index.html`□□□□□□□ CSS/JS□

□□□□

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```
/root/autodl-tmp/vision_qa_platform/results/accuracy/
modular_results_complex1000.json
```

□□□□□□□ `TRACE_RESULTS_FILE` □□□

□□ API

□□	□□
`GET /`	□□□□□
`GET /api/cases`	□□□□□□□□□□□□□□□□□□□
`GET /api/case/<qid>`	□□□□□□□□ + □□ URL
`GET /api/case/<qid>/image`	□□□□□□
`POST /api/ask`	□□□□ + □□□□□□□□□

□□□□□□□

`build_graph()` □JSON □□ `trace` □□ D3 □□ `nodes` + `links`□

```
□□□□
  ├─ □□□1 − □□1 / □□2 / □□3□□□□□□□
  ├─ □□□2 − ...
  └─ □□□3 − ...
  ├─ □□□□
  └─ □□□□□□□□□□□□
```

□□□□□`question` / `subq` / `candidate` / `candidate_selected` / `pred` / `gt`□□□□□□□□ □□□

□□□□□□□□

□□	□□	□□
□□□□ + □□□□	D3 □□□□□□□□□□□□□□□	□□□□ + □□□□□□□

□□□□□□□/□□□□——□□□□□□□□□□□□□□□□□□□□□□

□□7861□`chat_vqa_webapp` — □□□□ Demo

□□□□□□□□□□ + □□□□□□□□□□□□□□□□□□□□□

□□□□

- □□□`agent_workspace/chat_vqa_webapp/app.py`
- □□□`agent_workspace/chat_vqa_webapp/templates/index.html`

□ 7860 □□□

□□□	**7860 trace**	**7861 chat**
□□□□	□□□□□□	□□□□
□□	□	□
□□□□□	□□□□□□□	□
□□□	□□□□□□□□	□□□□□□

[illegible]	**7860 trace**	**7861 chat**
[illegible]	[illegible]	[illegible] + [illegible]

[illegible]7861 [illegible]

`chat_vqa_webapp/app.py` [illegible]

- `to_chinese_text()`[illegible]
- `to_chinese_answer()`[illegible]yes/no → [illegible]/[illegible] [illegible] + LLM[illegible]
- `localize_trace_to_chinese()`[illegible]
- `_TRANSLATION_CACHE`[illegible]

[illegible]

```
[illegible] + [illegible]
    → POST /api/ask
    → LLaVA [illegible]
    → [illegible]
    → [illegible]
    → [illegible] { answer, graph, trace, image_url }
    → [illegible] + [illegible] D3 [illegible]
```

[illegible] API

[illegible]	[illegible]
`GET /`	[illegible]
`GET /api/health`	[illegible]
`POST /api/ask`	[illegible] + [illegible]
`GET /uploads/<filename>`	[illegible]

[illegible]7862[illegible]chat_vqa_portal — [illegible]

[illegible] 7861 [illegible]

[illegible]

[illegible]	[illegible]
`chat_vqa_portal/app.py`	[illegible]Session[illegible]
`chat_vqa_portal/auth_utils.py`	[illegible]
`chat_vqa_portal/db.py`	MongoDB [illegible] Mongita [illegible] JSON[illegible]
`chat_vqa_portal/vqa_service.py`	[illegible] 7861 [illegible] + [illegible]
`chat_vqa_portal/templates/`	login / register / chat [illegible]
`chat_vqa_portal/static/`	CSS[illegible]JS[illegible]`chat.js`[illegible]

[illegible]

[illegible] `mongodb://127.0.0.1:27017`[illegible] `vision_qa_portal`[illegible] MongoDB [illegible] `mongo_data/`[illegible]Mongita[illegible]

[illegible]

[illegible]	[illegible]
`users`	username, email, password_hash, created_at
`conversations`	user_id, title, created_at, updated_at
`messages`	conversation_id, role, content, image_path, answer, **graph**, trace, visual_description

[illegible] **`graph`** [illegible] D3 [illegible]

[illegible]

- Flask `session + secret_key`[illegible] `PORTAL_SECRET_KEY`[illegible]
- `@login_required` [illegible] `/api/*` [illegible] 401[illegible] `/login`
- [illegible]`POST /api/auth/register`
- [illegible]`POST /api/auth/login`
- [illegible]`POST /api/auth/logout`

[illegible] API

[illegible]	[illegible]
`GET /api/conversations`	[illegible]
`POST /api/conversations`	[illegible]
`GET /api/conversations/<id>/messages`	[illegible]
`POST /api/conversations/<id>/ask`	[illegible]+[illegible]
`DELETE /api/conversations/<id>`	[illegible]
`GET /api/auth/me`	[illegible]

[illegible]GPT [illegible]

[illegible]	[illegible]	[illegible]
[illegible] + [illegible] + [illegible]	[illegible] + [illegible]	D3 [illegible] + [illegible]

`chat.js` [illegible]

1. [illegible] `loadConversations()` [illegible]
2. [illegible] → [illegible] `graph` [illegible]
3. [illegible] → `FormData` [illegible] → `/api/conversations/<id>/ask`
4. [illegible] + `renderGraph()`

[illegible]`uploads/<user_id>/<filename>`[illegible] `user_id == session`[illegible]

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□□	□□	□□
`PORTAL_PORT`	□□□□	`7862`
`MONGO_URI`	MongoDB □□□	`mongodb://127.0.0.1:27017`
`PORTAL_SECRET_KEY`	Flask □□□□	□□□□□□□□□□□□□

□□D3 □□□□□□□□□□□□□□

□□□□□□ **D3.js v7** □□□□□CDN□`https://cdn.jsdelivr.net/npm/d3@7`□□

□□□□□□

```
nodes = [{ id, label, type, detail }, ...]
links = [{ source, target }, ...]

d3.forceSimulation(nodes)
  .force("link", d3.forceLink(links).id(d => d.id))
  .force("charge", d3.forceManyBody())      // □□□□
  .force("center", d3.forceCenter(...))      // □□
  .force("collision", d3.forceCollide())    // □□□□7860/7861□

// □□ + □□□□ + □□ + □□□□ detail JSON
```

□□□□□□

□□	□□	□□
`question`	□□ #22d3ee	□□□
`subq`	□□ #f59e0b	□□□
`candidate`	□□ #60a5fa	□□□□□
`candidate_selected`	□□ #34d399	□□□□
`pred`	□□ #a78bfa	□□□□
`gt`	□□ #f43f5e	□□□□ 7860 □□□

□□

- □□□`d3.drag()` □□□□□□
- □□□□□□□□□□□ `detail` □□JSON□□□□□□□□□□□□□□□□

□□□□□□□□□

```
┌────────────────────────────────────────┐
│      □□□LLaVA + Qwen □□□□□            │
└────────────────────────────────────────┘
                   │
```

```
┌─────────────────────────────┼─────────────────────────────┐
        ▼
▼                             ▼
   trace_webapp                  chat_vqa_webapp
chat_vqa_portal
   (7860)                        (7861)
(7862)
        │
│                             │
   [illegible] JSON            [illegible]                   [illegible]
   + [illegible]             [illegible]                  + [illegible]
   [illegible]/[illegible]/[illegible]       [illegible]/[illegible]              + MongoDB [illegible]
   [illegible]           [illegible]           [illegible]/[illegible]/[illegible]
                                                                    [illegible]
[illegible]
```

[illegible]

- **7860** = [illegible]
- **7861** = [illegible] ChatGPT + [illegible]
- **7862** = [illegible] 7861 [illegible]

[illegible]

[illegible]	[illegible]
`agent_workspace/webapps_ctl.sh`	[illegible]
`agent_workspace/trace_webapp/app.py`	7860 [illegible]
`agent_workspace/chat_vqa_webapp/app.py`	7861 [illegible]
`agent_workspace/chat_vqa_portal/app.py`	7862 [illegible]
`reasoning_modules.py`	[illegible]
`tot_got_integration.py`	LLaVA + Qwen [illegible]
`vqa_eval_utils.py`	[illegible]
`results/accuracy/modular_results_complex1000.json`	7860 [illegible]

[illegible]2026-05-29